Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
2 P-only DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	**long**	0.6	1.43	1.89	8.35	0.827	0.517	0.985	5.68	0.971	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.271	0.75	18.7	0.82	0.466	1.21	0.49	0.985	1	0	Complete
4 T/P DeepONet10	**10k**	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	**long**	0.547	0.964	53.1	0.734	0.939	25.1	0.669	1.01	1	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.375	0.893	16.8	0.975	0.819	18.8	0.566	0.894	0.992	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.427	0.877	68.1	0.592	0.648	13.1	0.551	7.69	0.966	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.305	0.885	30.5	0.776	0.627	8.42	0.521	2.09	0.991	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.